

MTH316 Assessment: Dimensionality Reduction and Predictive Modelling for Air Quality Data

Topics Covered: Descriptive Statistics, Linear Regression, and Principal Component Analysis (PCA)

Due Date: [9:00am, 30 May, Saturday]

General Instructions

Read Carefully Before Starting

- **Dataset.** You are provided with 5 daily air quality datasets for 5 different locations in Beijing. The datasets contain the following six pollutant concentration as well as the temperature (TEMP) series measured over T consecutive days:

PM_{2.5}, PM₁₀, SO₂, NO₂, CO, O₃.

- You shall select **Only One dataset** for your own coursework. No need to repeat your work in multiple datasets. In your report, you shall clearly state your choice with the original dataset name.
- **Software.** Please use R as the programming language. Your R code shall start from reading the original raw dataset and all following processing procedures.
- **Submission.** Submit a single PDF report including your written answers, tables and figures. Include your R codes and the comments in the PDF report as Appendix.
- **Total marks:** 100.
- **Rounding.** Report numerical results to four decimal places.

Let $P_i(t, h)$ denote the observed **hourly** concentration of pollutant i at hour h on day t , where

$$i \in \{\text{PM}_{2.5}, \text{PM}_{10}, \text{SO}_2, \text{NO}_2, \text{CO}, \text{O}_3, \text{TEMP}\},$$

$h = 0, 1, \dots, 23$ represents the 24 hours within a day, and $t = 1, 2, \dots, T$ indexes the days.

For each pollutant i and day t , we use the **median** of the 24 hourly observations as the **daily** pollutant level, denoted by $P_i(t)$.

We then define the **daily proportional change** in pollutant i on day t as

$$r_i(t) = \frac{P_i(t) - P_i(t-1)}{P_i(t-1)}, \quad t = 2, 3, \dots, T.$$

Use **prior** value to interpolate if there is any Inf or NaN, or NA.

This transformation yields $n = T - 1$ observations for each of the seven variables. We collect these into the data matrix

$$\mathbf{R} = [\mathbf{r}_{\text{PM}_{2.5}} \ \mathbf{r}_{\text{PM}_{10}} \ \mathbf{r}_{\text{SO}_2} \ \mathbf{r}_{\text{NO}_2} \ \mathbf{r}_{\text{CO}} \ \mathbf{r}_{\text{O}_3} \ \mathbf{r}_{\text{TEMP}}] \in \mathbb{R}^{n \times 7},$$

where each column vector $\mathbf{r}_i \in \mathbb{R}^n$ contains the daily proportional changes for pollutant i .

1 Data Preparation and Descriptive Statistics (25 marks)

Q1.1 (15 marks) The original dataset contains hourly measurements for each pollutant i on every day. Using the February 2015 dataset of your choice, plot **one boxplot per day** based on the **24 hourly observations**, and plot all daily boxplots together in a **single figure** for pollutant i , in order to illustrate day-to-day variation across the month, where

$$i \in \{\text{PM}_{2.5}, \text{PM}_{10}, \text{SO}_2, \text{NO}_2, \text{CO}, \text{O}_3, \text{TEMP}\}.$$

In each figure, the **x-axis** should show the dates in **February 2015**, and the **y-axis** should show the corresponding pollutant values summarized by the daily boxplots. Produce **seven figures** in total: one for each variable listed above.

Finally, write a brief **2–3 sentence** comment describing any notable patterns observed in the boxplots across the pollutants during the month.

Q1.2 (10 marks) For each pollutant i and day t , we use the **median** of the 24 hourly observations as the daily pollutant level, denoted by $P_i(t)$. Compute the daily proportional change $r_i(t)$ for each of the seven pollutants. Produce a **summary statistics table** for the seven proportional-change variables over all available time periods. The table should include the following quantities for each variable:

Statistic	$r_{\text{PM}_{2.5}}$	$r_{\text{PM}_{10}}$	r_{SO_2}	r_{NO_2}	r_{CO}	r_{O_3}	r_{TEMP}
Count (n)							
Mean							
Std. Dev.							
Min							
25% (Q1)							
Median							
75% (Q3)							
Max							
Skewness							
Kurtosis							

Comment briefly (3–5 sentences) on any notable patterns: which pollutants exhibit the largest variability, whether any distributions appear highly skewed, and whether there are potential outliers.

2 Linear Regression for One-Day-Ahead Prediction of PM_{2.5} (35 marks)

In this section, you will build a regression model to predict the proportional change in PM_{2.5} one day ahead. Formally, define the response variable and the predictor variables as follows. All variables shall be calculated in question 1.

- **Response:** $y(t) = r_{\text{PM}_{2.5}}(t+1)$, the proportional change of PM_{2.5} on day $t+1$.
- **Predictors:** $r_{\text{PM}_{2.5}}(t)$, $r_{\text{PM}_{10}}(t)$, $r_{\text{SO}_2}(t)$, $r_{\text{NO}_2}(t)$, $r_{\text{CO}}(t)$, $r_{\text{O}_3}(t)$, $r_{\text{TEMP}}(t)$, i.e. all seven proportional changes observed on day t .

After alignment, you have $n-1$ usable observation pairs. Let $p=7$ denote the number of predictors.

Q2.1 (15 marks) Fit an ordinary least squares (OLS) multiple linear regression:

$$y(t) = \beta_0 + \beta_1 r_{\text{PM}_{2.5}}(t) + \beta_2 r_{\text{PM}_{10}}(t) + \beta_3 r_{\text{SO}_2}(t) + \beta_4 r_{\text{NO}_2}(t) + \beta_5 r_{\text{CO}}(t) + \beta_6 r_{\text{O}_3}(t) + \beta_7 r_{\text{TEMP}}(t) + \varepsilon(t).$$

Report the full regression output, including:

- The estimated coefficients $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_7$, their standard errors, t -statistics, and p -values.
- The coefficient of determination R^2 and the adjusted R^2 .
- The F -statistic for overall model significance and its p -value.

Your answer should include a table in the following format:

Table 1: OLS regression results for one-day-ahead prediction of PM_{2.5}

Variable	Coefficient	Std. Error	t -statistic	p -value
Intercept				
$r_{\text{PM}_{2.5}}(t)$				
$r_{\text{PM}_{10}}(t)$				
$r_{\text{SO}_2}(t)$				
$r_{\text{NO}_2}(t)$				
$r_{\text{CO}}(t)$				
$r_{\text{O}_3}(t)$				
$r_{\text{TEMP}}(t)$				
R^2				
Adjusted R^2				
F -statistic				
p -value for F -test				

Q2.2 (10 marks) Interpret the regression results:

- Which predictors are statistically significant at the 5% level? Provide a brief economic or scientific interpretation of the sign of each significant coefficient.
- Examine Pearson correlation coefficient matrix for all seven predictors. Are there signs of multicollinearity? Explain how multicollinearity could affect the reliability of individual coefficient estimates.

Q2.3 (10 marks) Evaluate model adequacy by performing the following diagnostics:

- (a) Plot the residuals versus fitted values. Comment on whether the assumptions of linearity appear reasonable.
- (b) Produce a Q-Q plot of the residuals. Comment on the normality assumption.

3 PCA-Based Regression (40 marks)

In this section, you apply Principal Component Analysis to the seven predictor variables and then use the resulting principal component scores as regressors.

Q3.1 (5 marks) Standardise (zero mean, unit variance) the seven predictor variables before performing PCA and show the summary statistics in a table as follows. Briefly explain why standardisation is important when the variables are measured on different scales.

Statistic	$r_{PM_{2.5}}$	$r_{PM_{10}}$	r_{SO_2}	r_{NO_2}	r_{CO}	r_{O_3}	r_{TEMP}
Count (n)							
Mean							
Std. Dev.							
Min							
25% (Q1)							
Median							
75% (Q3)							
Max							
Skewness							
Kurtosis							

Q3.2 (15 marks) Perform PCA on the standardised predictor matrix. Report:

- The eigenvalues $\hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \dots \geq \hat{\lambda}_7$ of the sample correlation matrix.
- The proportion of total variance explained by each principal component, and the cumulative proportion.
- A scree plot of the eigenvalues.
- Based on the scree plot and/or a cumulative variance threshold of approximately 80%, choose the number of components k to retain. Justify your choice.

Q3.3 (10 marks) Report the loading matrix $\mathbf{A}_k = [\hat{\mathbf{a}}_1 \ \hat{\mathbf{a}}_2 \ \dots \ \hat{\mathbf{a}}_k] \in \mathbb{R}^{6 \times k}$. For each retained component, describe in words which original pollutant variables contribute most strongly (i.e. have the largest absolute loadings) and suggest a possible interpretation of the component (e.g. “overall pollution level”, “photochemical vs. combustion contrast”, etc.).

Q3.4 (10 marks) Compute the principal component scores:

$$\mathbf{Y}_k = \mathbf{X}_c \mathbf{A}_k \in \mathbb{R}^{(n-1) \times k},$$

where $\mathbf{X}_c$ is the centred (and standardised) predictor matrix. Use these k PC scores as the new predictors and re-estimate the one-day-ahead regression:

$$y(t) = \gamma_0 + \gamma_1 \text{PC}_1(t) + \gamma_2 \text{PC}_2(t) + \dots + \gamma_k \text{PC}_k(t) + \eta(t).$$

Report:

- The estimated coefficients, their significance levels, and R^2 (adjusted R^2).
- Compare the R^2 of this PCA-based model with the full regression in Section 2. Has dimensionality reduction led to a meaningful loss of predictive power?

Summary of Marks

Section	Topic	Marks
1	Data Preparation and Descriptive Statistics	25
2	Linear Regression	35
3	PCA-Based Regression	40
Total		100

Marking Criteria

- **Correctness (40%):** Accurate computation of proportional changes, eigenvalues, loadings, communalities, regression coefficients, and diagnostics.
- **Interpretation and Discussion (30%):** Clear, concise explanations that demonstrate understanding of PCA and factor analysis theory (e.g. variance decomposition, orthogonality of PCs, rotational invariance of communalities, conceptual difference between PCA and FA).
- **Presentation (20%):** Well-organised report with properly labelled tables and figures, consistent notation, and professional formatting.
- **Code Quality (10%):** Readable, well- and detailed-commented R codes.